14	**D**	Displacement-time graph of SHM $a = a_0 \sin \omega t$ Velocity-time graph of SHM $v = \int a_0 \cos \omega t \, \mathrm{d}t = -\dfrac{a_0}{\omega} \cos \omega t + C$ Kinetic energy $E_\mathrm{k} = \dfrac{1}{2} m v^2$ $= \dfrac{1}{2} m \left[\dfrac{a_0}{\omega} \cos \omega t \right]^2$ $= \dfrac{1}{2} m \left(\dfrac{a_0}{\omega} \right)^2 \cos^2 \omega t$ which is described by the graph in Option D. Maximum kinetic energy

Qn	Ans	Solution
		$E_{k_{max}} = \frac{1}{2}m\left(\frac{a_0}{\omega}\right)^2$
15	A	The Barton's pendulums experiment demonstrates the physical phenomenon